

AI ASSISTED CODING WEEK-10.3

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B-27

Problem Statement 1: AI-Assisted Bug Detection

Scenario: A junior developer wrote the following Python function to calculate factorials:

Logic Error Code:

```
def factorial(n):  
    result = 1  
    for i in range(1, n):  
        result = result * i  
    return result  
print(factorial(5))
```

```
C:\Users\HARSHAVARDHAN\Desktop\AI Assitant Coding>C:\Users\HARSHAVARDHAN\AppData\Local\Programs  
\Python\Python314\python.exe "c:/Users/HARSHAVARDHAN/Desktop/AI Assitant Coding/fact.py"  
24
```

Corrected Code:

```
def factorial(n):  
    result = 1  
    for i in range(1, n + 1):  
        result = result * i  
    return result  
print(factorial(5))
```

```
C:\Users\HARSHAVARDHAN\Desktop\AI Assitant Coding>C:\Users\HARSHAVARDHAN\AppData\Local\Programs  
\Python\Python314\python.exe "c:/Users/HARSHAVARDHAN/Desktop/AI Assitant Coding/fact.py"  
120
```

Error Explanation:

- Bug: The loop used `range(1, n)` so `n` was excluded; the function computed $(n - 1)!$ instead of $n!$.
- Why: In Python, `range(a, b)` generates integers `a..b-1` (the stop is exclusive), so `range(1, n)` multiplies 1 through `n-1`.
- Fix applied: Use `for i in range(1, n + 1):` so the product includes `n`.
- Effect: In `fact.py`, `factorial(5)` now returns 120 (correct) instead of 24.

Problem Statement 2: Task 2 — Improving Readability & Documentation

Scenario: The following code works but is poorly written:

Logic Error Code:

```
def calc(a, b, c):  
    if c == "add":  
        return a + b  
    elif c == "sub":  
        return a - b  
    elif c == "mul":  
        return a * b  
    elif c == "div":  
  
print(calc(1, 2, "add"))  
print(calc(5, 3, "sub"))  
print(calc(4, 6, "mul"))  
print(calc(10, 2, "div"))
```

```
C:\Users\HARSHAVARDHAN\Desktop\AI Assitant Coding>C:\Users\HARSHAVARDHAN\AppData\Local\Programs  
\Python\Python314\python.exe "c:/Users/HARSHAVARDHAN/Desktop/AI Assitant Coding/fact.py"  
File "c:\Users\HARSHAVARDHAN\Desktop\AI Assitant Coding\fact.py", line 10  
    print(calc(1, 2, "add"))  
IndentationError: expected an indented block after 'elif' statement on line 8
```

Corrected Code:

```
def calc(a, b, c):  
    if c == "add":  
        return a + b  
    elif c == "sub":  
        return a - b  
    elif c == "mul":  
        return a * b  
    elif c == "div":  
        return a / b  
  
print(calc(1, 2, "add"))  
print(calc(5, 3, "sub"))  
print(calc(4, 6, "mul"))  
print(calc(10, 2, "div"))
```

```
C:\Users\HARSHAVARDHAN\Desktop\AI Assitant Coding>C:\Users\HARSHAVARDHAN\AppData\Local\Programs  
\Python\Python314\python.exe "c:/Users/HARSHAVARDHAN/Desktop/AI Assitant Coding/fact.py"  
3  
2  
24  
5.0
```

Error Explanation:

The error in the code is in the `elif c == "div":` block (line 8). **It's incomplete** — there's no return statement for the division operation.

When the "div" case is executed, the function doesn't return anything, so it implicitly returns `None`. This causes the last print statement: `print(calc(10, 2, "div"))` to output `None` instead of the division result.

Fix: Add the division operation to the div case: `elif c == "div": return a / b`

This will complete the function so all four operations (add, subtract, multiply, divide) work correctly.

Problem Statement 3: Enforcing Coding Standards

Scenario: A team project requires PEP8 compliance. A developer submits:

Logic Error Code:

```
def Checkprime(n):
    for i in range(2, n):
        if n % i == 0:
            return False
    return True

# AI-generated test cases
assert Checkprime(2) == True # 2 is a prime number
assert Checkprime(1) == True # 1 is not a prime number
print("All test cases passed!")
```

```
C:\Users\HARSHAVARDHAN\Desktop\AI Assitant Coding>C:\Users\HARSHAVARDHAN\AppData\Local\Programs
\Python\Python314\python.exe "c:/Users/HARSHAVARDHAN/Desktop/AI Assitant Coding/fact.py"
All test cases passed!
```

Corrected Code:

```
def check_prime(n):
    if n < 2:
        return False
    for i in range(2, n):
        if n % i == 0:
            return False
    return True

# AI-generated test cases
assert check_prime(2) == True # 2 is a prime number
assert check_prime(1) == False # 1 is not a prime number
print("All test cases passed!")
```

```
C:\Users\HARSHAVARDHAN\Desktop\AI Assitant Coding>C:\Users\HARSHAVARDHAN\AppData\Local\Programs
\Python\Python314\python.exe "c:/Users/HARSHAVARDHAN/Desktop/AI Assitant Coding/fact.py"
All test cases passed!
```

Error Explanation:

PEP8 issues in the submitted code:

1. Function name uses PascalCase instead of snake_case.
2. Missing indentation inside the function body.
3. Missing spaces around operators (%).
4. No blank line before function definition (in a larger file).
5. Function name does not describe behavior clearly per PEP8 conventions.
6. No handling of invalid inputs like numbers less than 2.

Problem Statement 4: AI as a Code Reviewer in Real Projects

Scenario: In a GitHub project, a teammate submits:

Original Code:

```
def processData(d):  
    return [x * 2 for x in d if x % 2 == 0]
```

Refactored Function Based on Feedback:

```
from typing import List, Union  
  
def multiply_even_numbers(  
    numbers: List[Union[int, float]],  
    multiplier: Union[int, float] = 2  
)-> List[Union[int, float]]:  
    if not isinstance(numbers, list):  
        raise TypeError("Input must be a list")  
  
    return [  
        num * multiplier  
        for num in numbers  
        if isinstance(num, (int, float)) and num % 2 == 0  
    ]
```

Explanation:

The function fails because it assumes the input is a list of integers without any validation. If the input is non-iterable or contains non-numeric elements, the `% 2` operation raises a `TypeError`. The unclear function and variable names also make the intended behavior hard to understand.

Problem Statement 5: — AI-Assisted Performance Optimization

Scenario: You are given a function that processes a list of integers, but it runs slowly on large datasets:

Given Code:

```
def sum_of_squares(numbers):  
    total = 0  
    for num in numbers:  
        total += num ** 2  
    return total  
  
print(sum_of_squares([1, 2, 3, 4])) # Output: 30
```

Optimized version:

```
def sum_of_squares_optimized(numbers):  
    return sum(x * x for x in numbers)  
  
print(sum_of_squares_optimized([1, 2, 3, 4, 5]))
```

Explanation: The function has a time complexity of $O(n)$ since it processes each element once, which makes it slow for very large lists. Using Python's built-in sum with a generator expression reduces loop overhead and improves performance. Both versions do the same work, but the optimized one runs faster due to better internal implementation. The trade-off here improves performance without hurting readability.